

Data : Ethics & GDPR

Ethics in Data Collection – Is it important?

- ▶ YES! We have seen over the past few lectures the amount of data that is being collected about us
- ▶ Data, Metadata and Information:
 - ▶ Data: Raw form of information without context
 - ▶ In the example we have the “raw” data 38
 - ▶ Metadata: Data about data, can give context
 - ▶ Example; metadata is that 38 is my favourite number
 - ▶ Information: Data given context or meaning
 - ▶ Combining the raw data 38 and the metadata we now know that 38 is my favourite number



Data, Metadata, and Information

More and more of what we do leaves a trace, and that trace is potentially valuable.

- Whether it is what Tik Tok videos you have liked, who you have followed on Instagram or what you have been searching for on amazon – all data when combined can become extremely valuable to those wishing to **exploit it – for good or bad!**

Data, Metadata, and Information

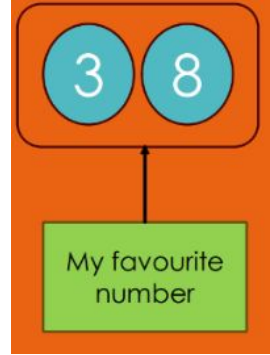
DATA:

Data, this is the raw form of information, without context. This can be 1 and 0 or just a string of characters. Here we have the “raw” data 38

METADATA:

Metadata, this is data about data, it can give it context.

This can be things like the time and date a file was created, or settings on a camera. Here we have the metadata - Amanda's favourite number (38)



Data, Metadata, and Information

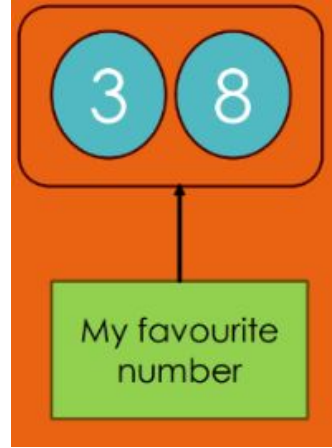
INFORMATION:

Information, is data given context or meaning.

So by combining the raw data 38 and the metadata we now know Amanda's favourite number is 38.

Traditionally data was data, and mostly useless without something like metadata to give it context. Then we started to enter the era of “big data.”

Suddenly there was mountains of data about even tiny, seemingly inconsequential things.



Case Study: Muslim Pro App

- ▶ A Lifestyle app that offers features and tools for Muslims to practice their faith.
- ▶ Over 150 million downloads worldwide.
- ▶ One feature is to capture your location and indicate the direction of Mecca. It did this with the seemingly obvious permission to access users' locations.
- ▶ In 2020, a report by Motherboard was released accusing Muslim Pro of providing data about its users to a third-party company called X-Mode.
- ▶ X-Mode then sold this data to various sources including the U.S. Military.

"The harvesting of data on Muslim app users worldwide is a serious threat to privacy and religious freedom." American Civil Liberties Union

Case Study : Muslim Pro App

- Created by Bitsmedia, Muslim Pro is said to be **the world's most popular Muslim lifestyle app**. It offers features and tools for Muslims to practice their faith.

For example, it provides users with **reminders of when to pray, and identifies your location so that it can indicate the direction of Mecca.**

- Bitsmedia say that the application is used by over **150 million people** worldwide
- Other apps, including a Muslim dating app were also named in the report.

Case Study : Muslim Pro App

- The ACLU filed a “Freedom of Information Request” for 3 years of records to investigate how the **U.S. government acquired this data** collected through apps used by millions of Muslims worldwide.

Muslim Pro, stopped sharing data with X-Mode after the story broke, but the report highlighted the unscrupulous people and companies that collect and distribute data and how it can end up being used by agencies without a direct connection to the apps themselves.

The Information is piling up...

- Metadata made data more useful
- Mountains of data collected - even tiny inconsequential things, when combined can form a detailed “profile” for a user
- These profiles started to be used for “targeted advertisements”
- People realised that data is valuable and who had the most data would control the digital landscape

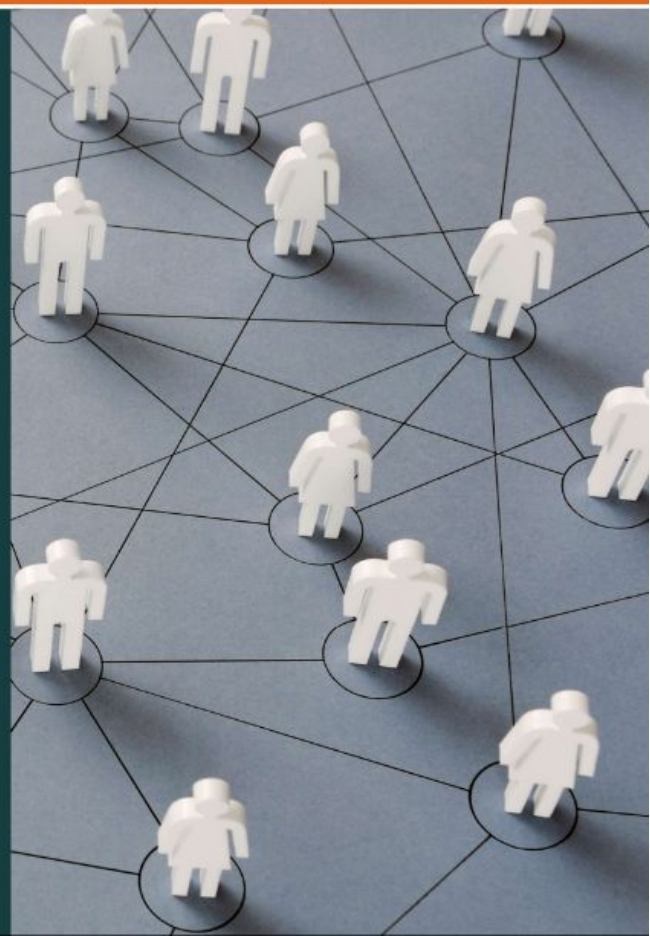
- For years it was a free for all with websites scraping as much data as it could about you;
 - Browser screen resolution, how you got to the page, search terms, what other websites you had visited, what you had been looking at on other websites...
- Even today; “tracking cookies” and “session ID tokens” are present on most sites you visit

Companies did not care what the data was

They just knew they had to get as much data as possible and make it into information later

Why collect Data when people just tell us....

- ▶ The rise of “Social Media”...
- ▶ People realised that we do not have to try and figure out information if we can just get people to tell us
- ▶ Facebook succeeded where others have failed:
 - ▶ Tracking every move you make in a browser whilst logged into facebook
 - ▶ Caught enabling the microphone and “listening” in the background
 - ▶ Tracking location in real time
 - ▶ Facebook are not the only ones...
- ▶ You can find out people’s likes, dislikes, shopping habits, who they are friends with, what they do and do not believe in



Targeted Data

- ▶ Led to a whole industry dedicated to obtaining as much data about people as possible, by any means, so that it could be sold on.
- ▶ Any company that stored data became a target, look at dark web sites selling stolen data.
- ▶ It's not always just credit card details, there are much more subtle ways data can be useful.
- ▶ Creating a realistic looking “bot” profile on a social media site is easy with enough training data, you can even start to emulate not just the way people look but how they use language.
- ▶ Don't believe me? A quick demonstration!

Data Mining: The Dark Side

- ▶ We have looked at the benefits of Data Mining in a previous lecture
- ▶ Can be used inappropriately to gather user data without the user's consent or for illicit purposes

EXAMPLE: Facebook and Cambridge Analytica

“App” asked users questions; collected data about the user and their friends

270,000 “used” the app → 85,000,000 users information

Data sold to political campaigns to create incredibly specific targeted adverts which influenced election results



- Even though data mining is a useful tool that can yield great results for businesses, it can also be used inappropriately if a business gathers user data without the user's consent or for illicit purposes.
- In 2013 an “app” developed by Dr Spectre (Now Dr Kogan) at Cambridge was designed to ask a series of questions and tell them information about their psychological profile.
- It was designed to also make use of the “Face Open Graph” platform, which allowed the app not just to collect data about its user, but also its user's friends.
- Only around 270,000 people “used” the app, but this allowed them to collect the information of over 85,000,000 users.
- This data was then mass harvested, and then **sold to political campaigns** who used it to create incredibly specific targeted adverts and messages designed to appeal to certain people.

Data Mining: The Dark Side

- ✓ Facebook claimed they collected: public profile, page likes, birthday and current city.
- ✓ Others claimed the app also had access to their news feed, timeline and messages
- ✓ Information collected from the app was enough to build psychological profiles of people → Create an add targeting a person's hopes or even fears
- ✓ Used extensively by the Trump campaign in 2016 and possibly Brexit
- ✓ 2018: whistle blower released a series of documents
- ✓ Multiple government investigations and raised wider concerns about privacy



Facebook claims the information collected "only" included: Public profile, page likes, birthday, and current city.

Some users also claim the app was given access to their news feed, timeline and messages.

Sources have claimed the information collected from the app was enough to build psychological profiles about people.

This would allow them to best create an add that could target a specific person hopes, or even fears. It was used to target users for misinformation campaigns.

It was confirmed to have been used extensively by the Trump campaign in 2016. And possibly even Brexit.

Eventually in 2018 a whistle-blower came forward to reveal what had been happening. Christopher Wylie who had previously worked for Cambridge Analytica released a series of documents about what the company had been doing. He said "They targeted people who were more prone to conspiratorial thinking. They used that data, and they used social media more broadly, to first identify those people, and then engage those people, and really begin to craft what, in my view, was an insurgency in the United States."

For a more detailed look at this see The Great Hack on Netflix.

Dark Patterns

- ▶ What are they?
 - ▶ Underhand and manipulative way of getting people to do what you want them to do on your company website.
- ▶ How?
 - ▶ Making Interface elements in such a way that one option is more “tempting”
 - ▶ Hiding terms and conditions behind pages of text
 - ▶ Promising a price on something, but then changing it at the last minute

A great piece of work at Princeton in 2019, surveyed 11,000 shopping websites and broadly categorised 7 “dark patterns” that were in common use.

Dark Patterns

Sneaking

- User is drawn into a transaction on false pretences, because important information is hidden or delayed from being presented to them.

Urgency

- User is pressured into completing an action because they are presented with a fake time limitation.

Misdirection

- Use of specific language or visuals to encourage users to make a specific choice.

Social Proof

- User is misled into believing a product is more popular or credible than it really is, because they were shown fake reviews, testimonials, or activity messages.

Scarcity

- User is pressured into completing an action because they are presented with a fake indication of limited supply or popularity.

Obstruction

- User is faced with barriers or hurdles, making it hard for them to complete their task or access information.

Forced Action

- User wants to do something, but they are required to do something else undesirable in return. Telling people, they must “sign up to X to continue.

Ethical Questions: Using Data Science...

Financial Institutions accept and deny bank loans or calculate insurance premiums



What is the social cost of a wrong decision for a bank loan or insurance...

Companies scanning CVs to recommend the best candidate



What is the chance for a bias towards gender or age if the algorithm uses past data...

Cookies monitoring browsing behaviour to target advertisements



Is this an intrusion of an individual's privacy...

Differential pricing by Airlines. Rideshare companies (e.g. Uber) use surge pricing based on demands.



Is there a risk of these companies exploiting individuals beyond their means when they are in desperate need of their services...

Airlines use data science to decide on differential pricing for individuals based on their needs.

Differential pricing: is a pricing strategy in which a company sets different prices for the same product on the basis of differing customer type, time of purchase etc. For example, airline companies use differential pricing and charge higher prices for business travellers and discounted prices for tourists.

Rideshare companies such as Uber use data for surge pricing.

Surge pricing: occurs when a company raises the price of its product or service if there is an increase in demand. FROM UBER: Bad weather, rush hour, special events may cause an increase in the number of people requesting an Uber at the same time. In these cases of high demand, the price may increase to ensure that those who need a driver can get one. This means that riders pay more, or they can wait a few minutes to see if the price goes down.

Ethical Concerns... Summary



Unfair Discrimination

Incorrect and unchecked use of data can lead to unfair discrimination against individuals; Gender, demographics, socio-economic conditions

If you have large datasets, you might not even realise that the data is biased - be aware of overtraining on certain characteristics

Reinforcing Human Biases

Using past data, generated by human decisions to predict future outcomes can lead to these biases being included in the result

Algorithms may be influenced by analysts' biases - they could choose data and hypotheses that are important to them

Ethical Concerns... Summary

Lack of Transparency

Model may predict an outcome but not explain the rationale behind the result

These “black box” solutions mean it is difficult for businesses to understand and explain the reason for a business decision

Privacy

Sensitive data stored by various organisations is subject to hacking and misuse

Example; the Cambridge Analytica data breach discussed earlier

Consent and Power

Organisations lack transparency in what data they are collecting; they use this to make decisions

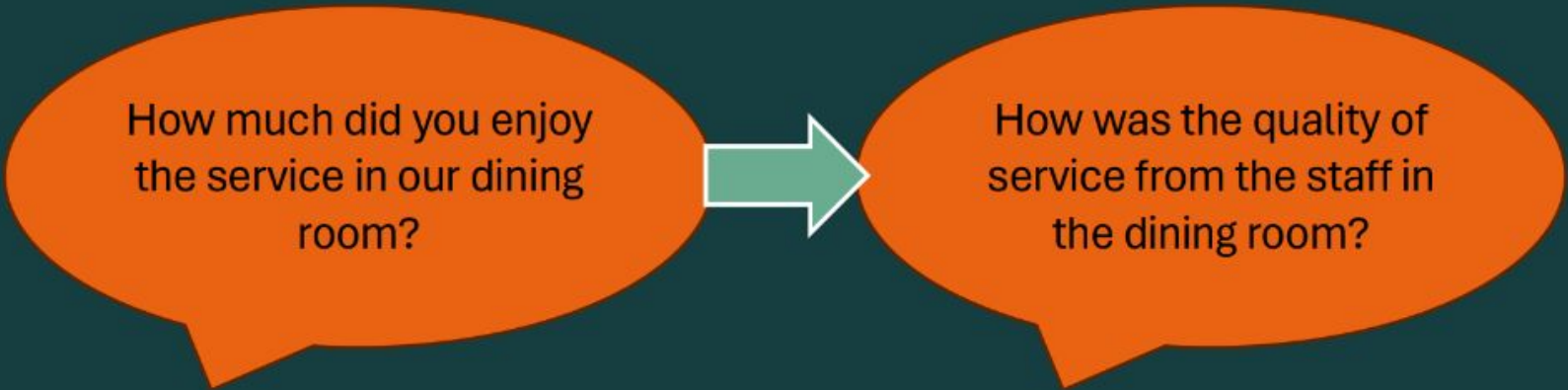
Browsers and Websites; Capture enormous amount of user data without their knowledge or consent

Data Analysis and Bias

Bias exists in both the Data itself and the people analysis or using it.

Adverse effects range from making bad decisions that affect profit to adversely affecting certain groups of people.

Human Sources: using unrepresentative data sets, leading questions in surveys, biased reporting and measurements



How much did you enjoy
the service in our dining
room?

The diagram consists of two orange speech bubbles connected by a green arrow pointing from left to right. The left bubble contains the text 'How much did you enjoy the service in our dining room?' and the right bubble contains the text 'How was the quality of service from the staff in the dining room?'.

How was the quality of
service from the staff in
the dining room?

Data Analysis and Bias

Propagating the Current State

► Amazon Recruitment Tool

- (2014) Automated tool to review job applications to make the recruitment process easier
- AI was trying to find recruits that matched existing workforce
- Trained by observing patterns on CVs over a 10-year period
- 2015 noticed it was not rating candidates in a gender-neutral way for software engineer roles
- Algorithm was biased when “looking” at candidates as the model matched on words more likely to appear on a male’s CV
- Recommended unqualified candidates based on how the model had been trained
- Project was scrapped as they could not guarantee the models not to be discriminatory in some other way



Data Analysis and Bias: Example

Under-Representing Populations

- ▶ Examples of bias; Gender, Race, Economic status
- ▶ Data used to train model does not account for all factors that go into decision making
- ▶ Can have tragic consequences; in medicine – failing to highlight differences in heart disease symptoms between men and women

Confirmation Bias

- ▶ Researches only choose data that supports their hypothesis
- ▶ Carry out research with a preconceived idea in mind; find statistics to support that notion
- ▶ Often found when evaluating results; if results confirm hypothesis, we don't question them further. If hypothesis not confirmed; re-evaluate the process, data or algorithms thinking there must be a mistake.

Examples

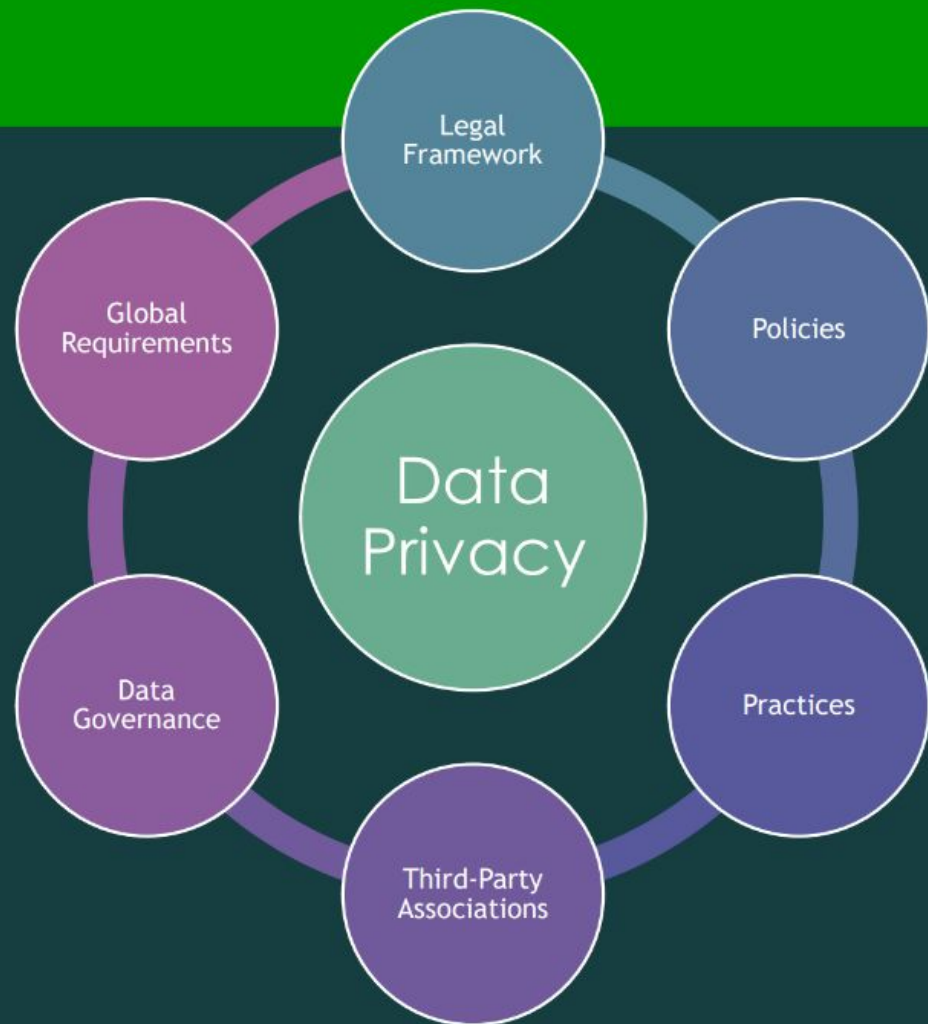
Political Views: When reading about politics on the internet, people tend to select articles that match their view.

Horoscopes: Horoscopes are deliberately written in a way that is slightly vague. If you analyse the statements carefully, you will discover that each one is open to a lot of interpretation. Events that happen afterwards could be construed in a way that in which it seems very consistent with what the horoscope predicted. True believers will always find a way to fit what happened to them with what their horoscope predicted. This is the magic of horoscopes and the power of the confirmation bias.

Personal Outlook: Being optimistic is good for a person's mental health, to some extent. Seeing the positive side of everything can keep us in a good mood. But optimists also seem to have a talent for ignoring negative or unpleasant information. Being pessimistic is just the opposite. Always seeing the negative side of a situation can make us feel down and lose a sense of hope. Both sides of this coin are good examples of confirmation bias. The optimist only looks for positive information and the pessimist only looks for negative information. The glass will always be half full or half empty, depending on your personal outlook.

Data Privacy

- Aspect of data protection that addresses; storage, access, retention, immutability and security
- Associated with proper handling of personal data or personally identifiable information (PII)
- Extends to other valuable or confidential data; financial, personal health information etc.
- Discipline using rules, practices, guidelines and tools to help organisations maintain privacy compliance.



Data Privacy Elements

Data privacy is not a single concept or approach. Instead, it's a discipline involving rules, practices, guidelines and tools to help organisations establish and maintain required levels of privacy compliance.

Data privacy is generally composed of the following **six elements**:

Legal framework:

Prevailing legislation enacted and applied to data issues, such as data privacy laws.

Policies: Established business rules and policies to protect employees and user data privacy.

Data Privacy Elements

Practices:

Best-practices put in place to guide IT infrastructure, data privacy and protection.

Third-party associations: Any third-party organisations, such as cloud service providers, that interact with data.

Data governance: Standards and practices used to store, secure, retain and access data.

Global requirements: Any differences or variations of data privacy and compliance requirements among legal jurisdictions around the world such as the U.S. and European Union (EU).

Why data privacy matters ?

Why do corporations collect so much data?

Identify Customers' Needs & Understand Infrastructure:

Using data to figure out market demands and optimize business workflows.

Glean Insights:

Mining data for business intelligence to make strategic decisions.

Train ML and AI Systems:

The peak value stage using aggregated customer data to build and train artificial intelligence and machine learning models.

The Vulnerabilities

It divides these **risks** into two distinct perspectives:

For the Individual:

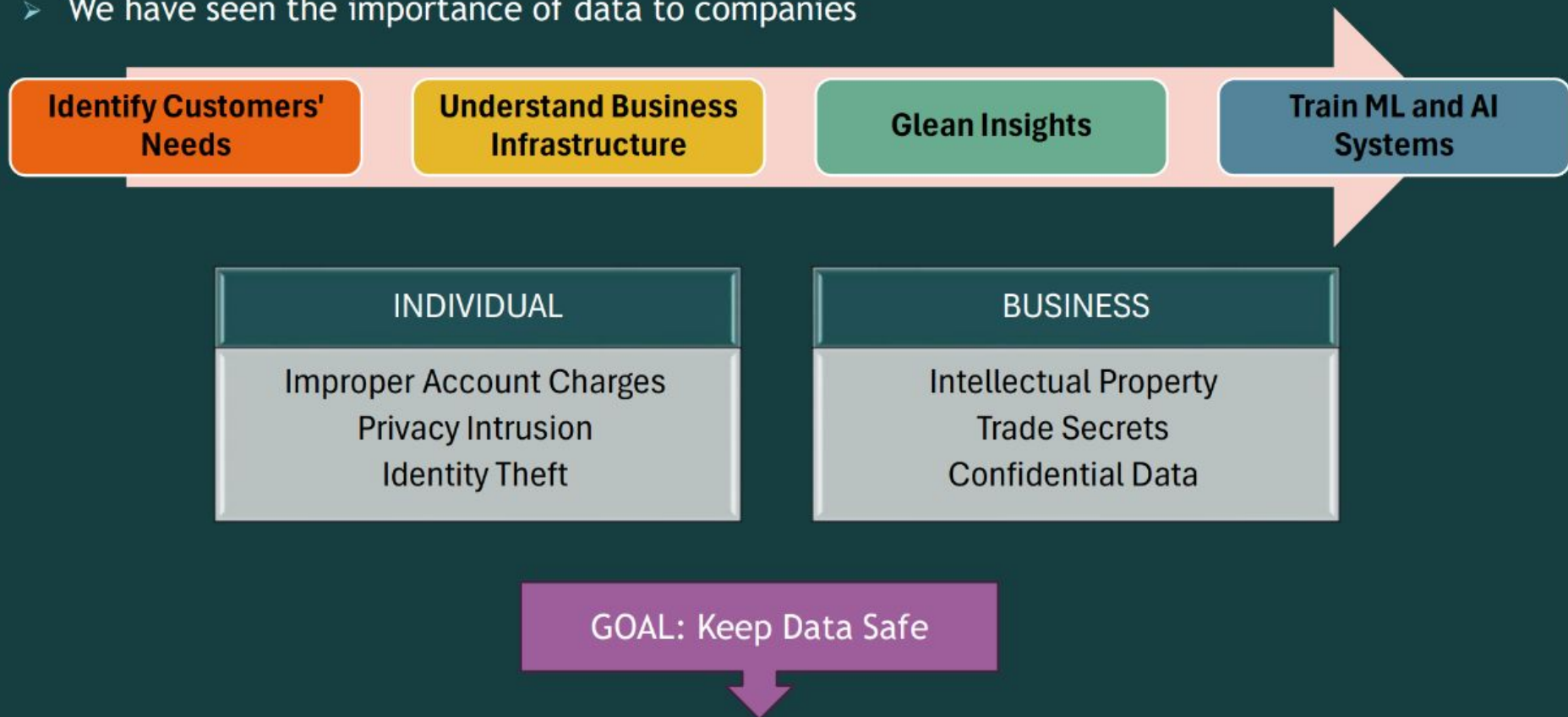
If data is compromised, human beings face personal devastation, such as financial fraud (Improper Account Charges), stalker-like corporate profiling (Privacy Intrusion), or digital impersonation (Identity Theft).

For the Business:

If a company leaks data, it faces commercial ruin, including losing its competitive edge (Intellectual Property & Trade Secrets) or legal/reputational damage from leaking corporate secrets (Confidential Data).

Data Privacy

- We have seen the importance of data to companies



The Importance of Data Privacy & Business Value

The importance of data privacy is directly related to the business value of data. The evolving data economy is driving businesses of all sizes to collect and store more data from more sources than ever before. Data is used for a range of business reasons, including the following:

- **to identify customers**, understand their needs and provide goods and services to them;
- **to understand the business infrastructure**, facilities and human behaviours based on data from networks and devices;
- **to glean insight from databases and data sources**; and
- to train machine learning and AI systems.

The Importance of Data Privacy & Business Value

Data privacy is a discipline intended to keep data safe against improper access, theft or loss. It's vital to keep data confidential and secure by exercising sound data management and preventing unauthorised access that might result in data loss, alteration or theft.

The Importance of Data Privacy & Business Value

For individuals, the exposure of personal data might lead to improper account charges, privacy intrusion or identity theft. For businesses, unauthorised access to sensitive data can expose intellectual property, trade secrets and confidential communications; it can also adversely affect the outcome of data analytics.

Data privacy lapses, also referred to as data breaches, can have a serious effect on all parties involved. Individuals affected by a data breach may find improper financial and credit activity in their name, compromised social media accounts and other issues. A business may face significant regulatory consequences, such as fines, lawsuits, and irreparable damage to their brand and reputation. With the integrity of its data compromised, a business may not be able to trust its data and need a response plan.

Data Protection

Vast majority of the provisions concerning data privacy are set out in 2 pieces of legislation:

- The Data Protection Act 2018
- The General Data Protection Regulation (GDPR)

Replaces previous Act from 1998

Concerned with processing of personal data; Data that helps identify an individual

Expressed in terms of principles and not explicit rules – Rules are often easy to get around, principles are often harder to break

All processing of personal data is covered with some exemptions; National Security, Crime & Taxation, Domestic Purposes

DATA
PROTECTION
ACT 2018

The Data Protection Act 2018 is the UK's implementation of the General Data Protection Regulation (GDPR).

Data Protection Act:

- **GDPR** and **The Data Protection Act** are two significant **legislations** that address the **protection and processing of personal data**.
- While GDPR provides a harmonised framework across the EU, the Data Protection Act tailors the regulations to the specific requirements of the UK.
- **The Data Protection Act 2018** controls how your personal information is used by organisations, businesses or the government.

Data Protection - GDPR

EU regulation enforced since May 25th 2018

GOAL: To standardise privacy laws across the European Union

Large fines for those found to have breached it; up to 20 million Euros (£17.5 million) or 4% of annual global turnover - whichever is greater

Individual member states were allowed to make small changes - The UK chose to update the Data Protection Act

GDPR is about protecting personal data that may allow a person to be identified, including pseudonymised data

GDPR - Terms

Personal Data

- Data that allows a living person to be directly or indirectly identifiable
- Sometimes this is obvious: Names, Location, Usernames
- Sometimes less so: IP Address, Cookie Data

This category of data is given extra protection

Sensitive Data

- Racial or Ethnic Origin
- Political Allegiance
- Religious Beliefs
- Trade Union Membership
- Biometric/Genetic Data
- Data around Sexual Orientation

GDPR - Terms

- ▶ Pseudonymised Data; Student ID, IP Address, Membership Number, Username

	Student Name	Student ID	Course of Study
Original Data	Joe Smith	12345	Computing
Pseudonymised Data	Candidate 1	Xxxxxxxx	Computing



Personal data which have undergone pseudonymisation, which could be attributed to a natural person by the use of additional information should be considered to be information on an identifiable natural person...”



GDPR – The Seven Principles

Lawfulness, Fairness and Transparency

- Data should be used lawfully, where local laws may apply.
- Data should be used fairly and in a transparent manner in relation to individuals.
- Regardless of individual characteristics.

Purpose Limitation

- Data should be collected for specific legitimate purposes.
- These should be explicitly stated.
- Data should not be used for uses incompatible with those purposes.
- One Exemption to this is for historical/statistical or scientific purposes.

Data Minimisation

- Data collected should be adequate, relevant and limited to what is necessary.

Accuracy

- Where possible, data should be kept up to date
- Where data has become inaccurate effort should be made to correct it or it should be erased.

Storage Limitation

- Data should be kept only for as long as it is necessary for the intended purposes.
- Except in cases of public interest or archival.

Integrity and Confidentiality (security)

- Data should be processed in a manner that ensures appropriate security of personal data.
- Including protection against Unauthorised access, Unlawful processing, Accidental loss, Destruction & Damage.

Accountability

- The controller shall be responsible for and be able to demonstrate compliance with the GDPR.
- This includes taking measures to be able to demonstrate compliance.

GDPR 7 main principles:

1. Lawfulness, fairness, transparency

Organizations must clearly explain:

- what data they collect
- why they collect it
- how it will be used

2. Purpose limitation

Data must only be used for the purpose it was collected.

Example: If you collect email for signup, you cannot use it later for unrelated marketing without permission.

GDPR 7 main principles:

3. Data minimization

Only collect what is necessary.

4. Accuracy

Data must be kept correct and updated.

5. Storage limitation

Do not store data longer than needed.

GDPR 7 main principles:

6. Integrity and confidentiality

Data must be secured against breaches or unauthorized access.

7. Accountability

Organizations must prove they comply with GDPR.

GDPR – Data Breaches

- ▶ Deliberate or accidental action (or inaction) by a controller or processor.
 - ▶ Such as mass deleting data.
- ▶ Sending personal data to an incorrect recipient.
 - ▶ Such as an accidental reply to all.
- ▶ Computing devices containing personal data being lost or stolen.
- ▶ Alteration of personal data without permission.
- ▶ Loss of availability of personal data.

GDPR – Data Breaches

In the event of
a Data Breach

- ✓ The relevant supervisory authority must be notified within 72 hours of you becoming aware of certain types of breach.
- ✓ If it is likely to affect an individual's rights and freedoms, you must also contact them “without undue delay”
- ✓ You should have mechanisms in place to detect, investigate and report breaches.
- ✓ A record should be kept of all breaches, even if notification is not required.
- ✓ For serious breaches, companies can be fined up to £17.5 million or 4% of their annual global turnover

Beyond the GDPR

Shows how the global conversation is shifting from **just regulating data privacy** to establishing absolute digital ownership and financial compensation for personal data.

Beyond the GDPR

Right to Be Forgotten

- Complete ownership of data and a true right to have your information completely deleted
- European Union → “Right to Erasure”
- Concept has existed for a many years: UK Rehabilitation of Offenders Act, 1974

Data Dividend Project

- Andrew Yang, movement dedicated to helping Americans regain ownership and control of their personal data
- Companies should pay to use your data, should you wish to share it

Beyond the GDPR

1. The Right to Be Forgotten (Shifting from Regulation to Erasure)

The Idea: True ownership means having the power to completely undo your digital footprint.

Under the EU's GDPR, this is officially codified as the "**Right to Erasure.**"

it draws a parallel to the UK Rehabilitation of Offenders Act of 1974, which legally allowed **certain past mistakes (criminal records) to be wiped** or "forgotten" so people could move on with their lives.

Beyond the GDPR

2. The Data Dividend Project (Shifting from Ownership to Monetization)

The Idea:

Personal data is a highly valuable commodity, and if tech companies are making billions from it, **the creators of that data (the users) should get a cut.**

Beyond the GDPR

Open Data Institute

- Founded by Sir Tim Berners-Lee in 2012
- Mission Statement: about working with organisations, businesses, governments and society; to build data infrastructure, knowledge, and strategies that create trust in data and data practices
- Lots of great resources on their website!

Open Data Institute (ODI) :

Data shouldn't just be locked away in corporate silos or heavily restricted by privacy laws.

Instead, it should be treated like physical infrastructure such as roads or electrical grids that society needs to function and innovate.

- ➔ By building robust data infrastructure, governments and businesses can share open, anonymized data (like weather, public transit, or health trends) to solve massive global problems.



SUMMARY

- ✓ Ethics; Data Usage and Analysis
- ✓ Avoiding Bias in Data Analysis
- ✓ Data Privacy
- ✓ Data Protection
- ✓ GDPR